SecAuthUAV: A Novel Authentication Scheme for UAV-Base Station Scenario

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***Abstract*—Drones, commonly known as Unmanned Aerial Vehicles (UAVs) are becoming very popular nowadays due to their wide applicability in areas ranging from package delivery systems to automated military applications. Nevertheless, communication security between a UAV and its base station (BS) is critical for completing its task without leaking sensitive information either to the adversaries or to unauthenticated users. UAVs are especially vulnerable to physical and node tampering attacks. Further, since UAV devices are generally equipped with small batteries and limited memory storage, lightweight security techniques are best suited for them. Addressing these issues, a lightweight mutual authentication scheme based on Physical Unclonable Function (PUF) for UAV-BS authentication is presented in this paper. We present a formal security analysis as well as old-fashioned cryptanalysis to show that our protocol provides various security features such as mutual authentication, user anonymity, etc. and is resilient against many security attacks such as masquerade, replay, node tampering attacks, etc. We also show a performance comparison of our protocol with the state-of-the-art UAV-BS authentication protocols, based on computation, communication and memory storage cost.**

***Index Terms*—UAVs, physical security, mutual authentication, security protocol, privacy, PUFs.**

I. INTRODUCTION

Around the world, Unmanned Aerial Vehicle (UAV) technology is being developed and deployed at a very rapid pace. From originally being used in military applications, these tools are being adopted across the government and corporate entities for providing a wide range of services including traffic management, goods delivery, inventory tracking, etc. With many governments easing regulations for UAV usage, the interest in this emerging technology is growing leaps and bounds leading to newer fields of applications being developed. UAVs can now can be used for traffic surveillance in the next generation Intelligent Transportation Systems [1], [2].

Although the UAVs provide various benefits, due to being deployed in the open environment they are also prone to several security threats. Primarily, the threats to UAVs come in two types- natural phenomena, and those caused by human interference [3]. The quality of data transmitted is influenced by natural phenomena, whereas human interference affects the security aspect of the transmitted data. Some of these security threats could be UAV node tampering attack, main in the middle attack, replay attack, etc. Communication of the UAVs with the base station being wireless in nature is thus prone to several of the above malicious attacks. For example, in a military surveillance scenario, if an adversary UAV tries to impersonate itself as a trusted UAV and authenticates itself to the base station, it can get secret information about the trusted UAV from the base station or send false information to the base station. This can cause serious disruptions to its services, leading to major economic or even human loses.

Node authentication is the first security aspect of any IoT network, which should be met before beginning a secure communication session in the network. Internet of Drones (IoD) is an IoT network which is implemented using drones or UAVs as the IoT devices [4]. The existing Internet security techniques implement authentication using stored cryptography keys. UAVs being deployed in the open air are prone to device capture attack, and thus to secret keys being exposed. Further, these UAVs generally have limited memory and computation capability, thus storing the secret keys and executing standard cryptography algorithms is also a challenge. This paper addresses the computation capability and the problems associated with the generation and storage of secret keys, by presenting a light-weight mutual authentication protocol for authenticating UAVs with a Base Station (BS).

In several recent studies, Physical Unclonable Functions (PUFs) have been shown to be very promising in providing security in a number of IoT and embedded system based applications [5–7]. Due to the inherent randomness introduced during the manufacturing process of these devices, they provide unclonable and unique identities to their devices. Thus they are suitable to be employed as a hardware root of trust and as hardware security primitives. The operation of PUFs is based on challenge-response pair mechanism, i.e. PUF has a property that when an input stimulus called the challenge is applied to it, the device responds with an output called the response. Two different PUF devices manufactured using the same fabrication process and with the same configuration are expected to produce different responses given the same challenge. In the proposed protocol, the above property of PUF is exploited for generating a unique session key for each device which is embedded with a PUF. By using PUFs in an interconnected system such as IoT or UAV network, every single device can have its unique fingerprint which cannot be cloned or reproduced. A PUF behaves like a mathematical function whose input (challenge) and output (response) are both in the form of a string of bits, as represented here:

$$R = PUF(C) \tag{1}$$

where the challenge C is given as input and the response R

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is the corresponding output to that challenge.

The major contributions of this paper are as follows:

i. We propose a PUF-based security protocol, SecAuthUAV which is capable of establishing a unique session key between the UAV and BS, after each round of authentication, without storing any secret information on the UAV.
ii. SecAuthUAV achieves mutual authentication, data confidentiality, identity protection, message integrity, and physical security and also ensures protection against replay, MITM, impersonation, and node-tampering attacks.
iii. We provide a formal security proof using Mao and Boyd logic, and also perform cryptanalysis to guarantee security and versatility of the protocol.
iv. A comparison of SecAuthUAV with other state-of-the-art UAV-BS security protocols, in terms of security features, computational cost, communication cost and storage cost is also provided.

The organization of the rest of the paper is as follows. Section II discusses the related works in securing UAV networks. The system model, security goals, attack model and assumptions for our system model are discussed in Section III. In Section IV, we present our mutual authentication protocol (SecAuthUAV) based on PUFs. In Section V, we subject our protocol to formal security analysis. Performance analysis of our protocol and comparison with other state-of-the-art security protocols for UAV-BS authentication is shown in Section VI. We finally conclude the paper in Section VII.

II. Related Works

UAV Networks (UAVNs) differ greatly from the other traditional wireless networks like Mobile Ad-Hoc Networks (MANETs) due to greater node mobility and thus the existing security solutions for MANETs and other traditional wireless networks cannot be adapted to UAVNs. He et al. have explained the essential features of UAV communications and identified the requirements for UAV security protocols in [8]. They have shown that traditional security techniques such as anomaly detection induce significant delays and therefore cannot be used for time-critical applications like those in UAVNs. Thus, they establish the need for UAV security protocols which take into account the trade-off between network performance and security strength.

Hooper et al. have presented a multi-layer framework in [9] for defending commercial UAV systems from Address Resolution Protocol (ARP) cache poisoning, buffer overflow and Denial of Service (DoS) attacks. Blazy et al. have considered a strong adversarial model which can perform fault injection, side-channel and physical attacks on a UAV [10**?**]. In their scheme, they store a single initial key which is used to generate subsequent keys using a keyed hash function. They propose to use multiple streams of such key sequences to prevent an attack. But, if a powerful attacker gains just one key of each stream, he/she could very easily keep track of all the key streams simultaneously thereby rendering all further communication insecure.

Several authentication and other cryptographic frameworks have been proposed for UAV and IoD networks in recent times. Chen et al. have proposed the Direct Anonymous Attestation with Mutual Authentication (MA-DAA) scheme in [11] for use in network-connected UAV systems. Compared with the existing Direct Anonymous Attestation (DAA) schemes, MA-DAA scheme is well suited to the low bandwidth and computation capabilities of the UAV networks. However their solution is based on Trusted Platform Modules (TPMs), which are specialized expensive security co-processors that need to be integrated into the systems leading to higher costs. Abdallah et al. have presented a lightweight security scheme for surveillance UAV networks in [12], however their scheme ensures only one-sided authentication, and not mutual authentication. Another work [13] is a certificateless group key authentication protocol targeted at untrusted UAV networks. This work is based on bilinear pairing and elliptical curve cryptography (ECC) and thus is not lightweight. Another work [14] proposes a short certificate based signature distribution scheme which is also heavyweight due to usage of public key cryptographic techniques. Two more works have been proposed by authors in recent times [15, 16] which have been proven to be sufficiently lightweight and secure against multiple attacks in IoD deployment scenarios. However, they are not secure against UAV node tampering and physical attacks. Since, secret information required for authentication is stored in the UAV's physical memory, an adversary can easily extract critical information and launch various attacks.

PUFs have been shown to be very effective in securing wireless and unmanned IoT environments recently [6, 17]. Chen and Willems propose a secret key generation protocol based on PUF in [18]. In another work [19], researchers have shown the feasibility of effectively extracting the security keys using light weight fuzzy extractors well suited to the IoT environment. The proposed protocol in this paper makes use of this concept to do away with the storage of critical information such as secret keys in the device's memory. Instead it supports generation of secret information on the fly based on the response generated from the inbuilt PUFs. Using this generated information, the proposed protocol authenticates the UAV node and generates a secure session key for further communication between the UAV and the base station. Thus, PUF gives the proposed protocol an edge over other protocols in terms of UAV physical security against node capture attacks.

III. System & Threat Model

A. System Model

The system model used in this paper consists of three types of entities namely trusted UAVs, malicious UAVs, and a base station (BS) as shown in Fig. 1. UAVs have limited memory and computational capability as compared to the BS. In this model, multiple UAVs may be connected to a single BS. The objective of this paper is to establish a session key between a UAV and the base station. Following this session key establishment, the UAV and BS can communicate data securely with each other. However, our focus is only on the session key establishment and not on the security of the subsequent data communication.

The computational device on every UAV is equipped with a PUF, which is used for generating a 32-bit response output

for a 32-bit challenge input to it. The UAV then performs a key expansion on this generated response to produce multiple subkeys. These subkeys are further used in the authentication protocol as discussed in Section IV. Both the UAV and the BS are capable of performing this mutually agreed-upon key expansion procedure. The registration of a UAV with the BS is carried out at a secure facility/location. Here, a challenge-response pair (C, R) of the UAV's PUF is extracted and securely added to the BS's database. Now, this UAV is ready to be deployed. Further in this model, for each UAV the BS starts with a single initial (C, R) pair which was saved at the time of UAV registration. For every instance of UAV's authentication with the BS, its PUF is used to generate a new challenge-response pair (C', R') pair on the fly. This new pair (C', R') is securely transmitted to the BS as can be seen in the last message being sent by the UAV in the protocol diagram in Fig. 2. Upon receiving the message, the BS retrieves the new (C', R') pair from the message and stores in its database to replace the existing (C, R) pair for the respective UAV. The ID of the authenticated UAV is dynamically changed at the end of the authentication both in the BS and in the UAV. This new ID is used for subsequent communication until it is changed at the end of next authentication.

Table I lists the notations used in this paper and their descriptions.

B. Attack Model

An untrusted UAV may try to authenticate with the BS by impersonating as a trusted UAV or man in middle attack. Since the communication channel is wireless and public, an intruder may try to eavesdrop on the transmitted messages or modify these messages. Moreover, in this model, an adversary may also have the capability to physically capture a flying UAV and extract secret information from its memory. So our model takes in account the standard attack model used in papers such as [**?**]

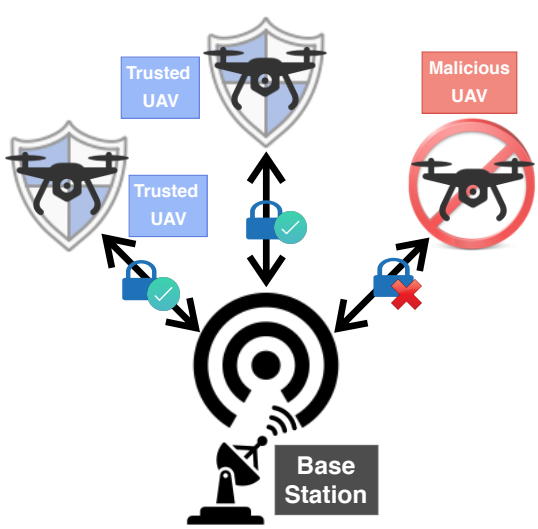


Fig. 1: System model

C. Security Goals

Keeping the following security goals in view, this protocol has been proposed.

i. Achieving mutual authentication between the individual UAV nodes and the BS.
ii. The communication between the UAV and the BS must be confidential.
iii. The protocol must be capable of generating a unique session key for each authentication session.
iv. The session keys generated must be safe against physical attacks like UAV node capture and node tampering.
v. If the communicated messages are tampered, the receiver, either BS or UAV, must be able to detect the tampering, and infer the presence of an attacker.
vi. No unauthorized entity must be capable of tracking the ID of a UAV.

D. Assumptions

We discuss here a few assumptions that are made in this paper.

i. Each UAV is endowed with a unique PUF. In case of a UAV capture, any attempt to tamper with the PUF will render the PUF unusable and the captured UAV will not be able to authenticate with the BS.
ii. UAVs have limited computational power and are memory constrained while the BS has no such constraints.
iii. The UAV itself does not store any secret information, rather relies on the response R generated using the PUF to be used as a secret to produce m subkeys to be used in the authentication protocol.
iv. Upon successful authentication, a secure session key is established between the UAV and the BS. This session key will be used for the next data communication session.
v. A session here refers to the actual data communication session between a trusted UAV and the base station. This paper does not focus on this data communication session. Rather it focuses on the authentication protocol to take place prior to this session, and the establishment of a secure session key to be used in this session.
vi. Every new UAV has to be first registered with the base station, before it can be successfully authenticated. Upon registration, its ID U_i and an initial challenge-response pair (C, R) generated from its PUF are saved in the base station's database.

IV. Proposed Drone Authentication (SecAuthUAV) Protocol

This section presents the proposed SecAuthUAV authentication protocol between a UAV and the BS which is depicted in Fig. 2. This protocol ensures that only trusted UAVs can authenticate with the BS before they can begin a communication channel with the BS. In this process of authentication, a secure session key is established between the UAV and the BS, which can be used for further communication.

When a UAV U_1 wants to authenticate with a nearby BS B_1, it sends its ID U_1 and a nonce N_A to B_1. B_1 checks within its database if U_1 exists and whether nonce

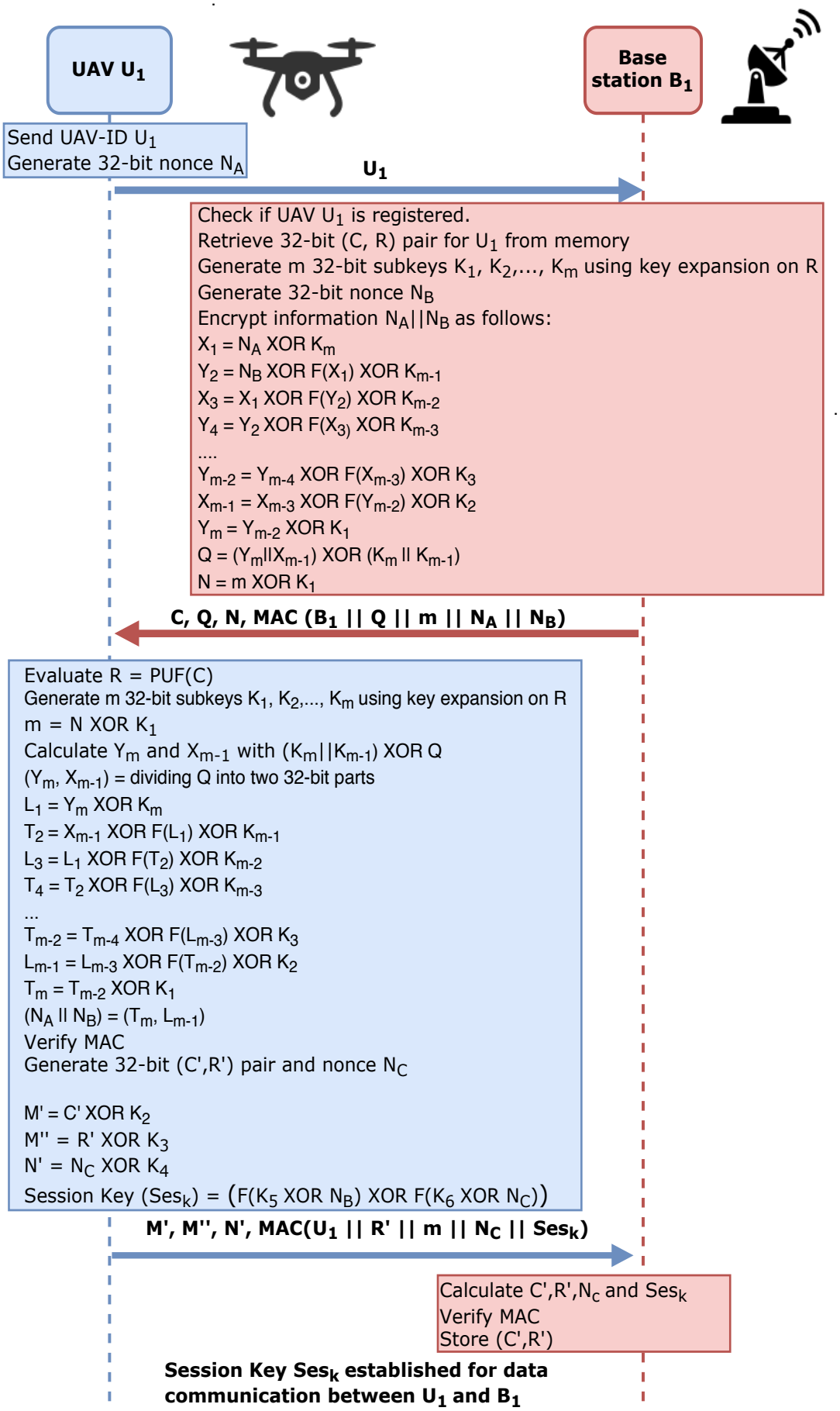


Fig. 2: SecAuthUAV protocol between UAV and the Base Station

TABLE I: Notations used in the paper

Notation	Description
U_i	ID of ith UAV
B_i	ID of ith BS
$\|$	Concatenation
$\oplus$	XOR operation
$MAC()$	Message Authentication Code
N_A, N_B, N_C	Nonces generated
Q	Cipher text prepared by BS
$F()$	Any public non-linear function with 32-bit input and output
$H()$	Hash function

N_A is fresh, i.e, it should not match the N_A generated in the previous authentication. If either of the condition fails, the authentication request by U_1 is terminated. Using U_1, the base station B_1 finds the corresponding 32-bit challenge-response pair (C, R) from its database. Using a known public key expansion procedure [20], it generates m 32-bit keys called $K_1, K_2, K_3, ..., K_m$ from the response R of the challenge-response pair (C, R). It must be mutually agreed upon by both the BS and the UAV. For example, the key expansion procedure used in AES or Blowfish encryption algorithm can be used here, which generates multiple subkeys to be used further in the protocol. The value of m could be any even number greater than 2, which can be decided upon based on the speed of the algorithm required. The base station B_1 then generates a 32-bit nonce N_B, which along with N_A is the information to be encrypted, as $N_A||N_B$.

$$\begin{gathered} X_1 = N_A \oplus K_m \\ Y_2 = N_B \oplus F(X_1) \oplus K_{m-1} \\ X_3 = X_1 \oplus F(Y_2) \oplus K_{m-2} \\ Y_4 = Y_2 \oplus F(X_3) \oplus K_{m-3} \\ ... \\ Y_{m-2} = Y_{m-4} \oplus F(X_{m-3}) \oplus K_3 \\ X_{m-1} = X_{m-3} \oplus F(Y_{m-2}) \oplus K_2 \\ Y_m = Y_{m-2} \oplus K_1 \end{gathered}$$

The final cipher-text Q to be sent to U_1 and the value m is encoded in the following way.

$$\begin{gathered} Q = (Y_m \| X_{m-1}) \oplus (K_m \| K_{m-1}) \\ N = m \oplus K_1 \end{gathered}$$

B_1 then sends C, Q, N and a Message Authentication Code $MAC(B_1 \| Q \| m \| N_A \| N_B)$ to U_1. Within the MAC, the first parameter verifies the identity of the BS, while data integrity is ensured by the second and third parameters. Freshness of the source (B_1 in this case) is identified by the last two parameter N_A and N_B. On receiving the message from the BS, U_1 generates the response R by passing the received challenge C through the PUF as given in (2). Using the same key generation and expansion procedure as used by the BS, it regenerates m 32-bit keys $K_1, K_2, K_3, ..., K_m$ from the response R. It calculates m and $Y_m \| X_{m-1}$ as shown below.

$$R = PUF(C) \tag{2}$$

$$m = N \oplus K_1 \tag{3}$$

$$(Y_m \| X_{m-1}) = (K_m \| K_{m-1}) \oplus Q \tag{4}$$

This is divided into two 32-bit strings to get Y_m and X_{m-1}. The decryption procedure is identical to the encryption procedure as can be seen in the following set of equations.

$$\begin{gathered} L_1 = Y_m \oplus K_m \\ T_2 = X_{m-1} \oplus F(L_1) \oplus K_{m-1} \\ L_3 = L_1 \oplus F(T_2) \oplus K_{m-2} \\ T_4 = T_2 \oplus F(L_3) \oplus K_{m-3} \\ ... \\ T_{m-2} = T_{m-4} \oplus F(L_{m-3}) \oplus K_3 \\ L_{m-1} = L_{m-3} \oplus F(T_{m-2}) \oplus K_2 \\ T_m = T_{m-2} \oplus K_1 \end{gathered}$$

Now, the original information $N_A||N_B$ is recovered as $T_m \| L_{m-1}$. From this, N_B is retrieved and the MAC is recomputed. Thus, the source of the message, its freshness and integrity are verified. In case MAC verification does not succeed, B_1 terminates the authentication of U_1. If verified, a new 32-bit challenge-response pair (C', R') is generated by U_1 using its PUF. It also generates a new Nonce N_C. This newly generated information is encoded into $M', M''andN'$ as follows.

$$\begin{gathered} M' = C' \oplus K_2 \\ M'' = R' \oplus K_3 \\ N' = N'_C \oplus K_4 \end{gathered}$$

The session key Ses_k with which further communication will take place is computed as shown below.

$$Ses_k = ((F(K_5 \oplus N_B) \oplus F(K_6 \oplus N_C)) \tag{5}$$

M', M'', N' and a new MAC with five parameters is sent to B_1. B_1 calculates C', R', N_C and Ses_k as shown in the following equations.

$$\begin{gathered} C' = M' \oplus K_2 \\ R' = M'' \oplus K_3 \\ N'_C = N' \oplus K_4 \\ Ses_k = ((F(K_5 \oplus N_B) \oplus F(K_6 \oplus N_C)) \end{gathered}$$

Now, B_1 has all the parameters to verify MAC. If the verification fails, authentication is terminated by B_1. If verified, it stores the new 32-bit challenge-response pair (C', R') for UAV U_1 in its memory replacing the previous entry (C, R). Until U_1 moves out of B_1's range, their communication is encrypted using Ses_k. Thus, mutual authentication has been achieved. Both U_1 and B_1 compute the new Alias ID (AID) for U_1 and update it in their storage.

$$AID = H(U_1 || K_7) \tag{6}$$

Changing the ID of the UAV using a suitable hash function as shown in equation (6) ensures untraceability. An adversary will not be able to match the AID with any previous IDs.

V. Security Analysis

To evaluate the security of the proposed protocol, we provide a formal security proof in section V-A, as well as old-fashioned cryptanalysis V-B. Such an approach is necessary because a formal proof by itself is insufficient for assuring actual security [21, 22], and it guarantees security and versatility of a security protocol. But, several attack-oriented goals can be traced to two formal security features, namely semantic security and mutual authentication [23]. Formal verification of the protocol guarantees that an adversary cannot gain or modify any information communicated in the network. We have used Mao Boyd logic [24] for formal verification of our protocol's security. Our old-fashioned cryptanalysis is based on 12 evaluation criteria (C1 - C12) as used by the authors in [25].

A. Formal Proof

In our proof we invoke the ***authentication***, ***nonce-verification***, ***confidentiality***, ***super-principal***, ***intuitive*** **and** ***good-key inference rules*** of Mao Boyd logic. Throughout the verification process, we denote the UAV U_1 by U and the Base station B_1 by B.

We first prove the statement "U believes N_B is a good secret between U and B". The following proof is shown pictorially in Fig. 3a. The (C, R) pair of U is stored in B, therefore it can be said that U believes R is a good secret between U and B. This is written in Mao Boyd logic as shown in (i). Using R, the UAV U is able to decipher Q in message 2 of the protocol to get $N_A \parallel N_B$. Since U is aware that N_A is contained in the first 32 bits, it can extract N_A. Therefore, we have the statement "U sees N_A with decipher key R" which is written symbolically in (ii).

$$U \models U \overset{R}{\leftrightarrow} B \tag{i}$$

$$U \overset{R}{\lhd} N_A \tag{ii}$$

Applying the ***authentication rule*** in [24] to statements (i) and (ii), we arrive at the statement "U believes B encrypted N_A using R" which is statement (iii). Since U generates a new nonce N_A each time, we have the statement "U believes N_A is fresh" which is (iv). Next, we apply the ***nonce-verification rule*** to statements (iii) and (iv) to get (v) which is "U believes that B believes that R is a good secret between U and B".

$$U \models B \overset{R}{\mid\!\sim} N_A \tag{iii}$$

$$U \models \#(N_A) \tag{iv}$$

$$U \models B \models U \overset{R}{\leftrightarrow} B \tag{v}$$

Using R, the UAV U decrypts Q and acquires N_B in message 2 of the protocol. Therefore, we next have the statement "U sees N_B with decipher key R" which is (vi). Applying the ***authentication rule*** to (i) and (vi), we get (vii) which is "U believes B encrypted N_B using R". Since B is known to generate a new nonce N_B each time, U is aware of the fact that no one other than B could have seen N_B. Thus it can be said that "U believes that B believes that no one other than U has access to N_B" which is (viii). Applying the ***confidentiality rule*** to (v), (vii) and (viii) we get (ix) which states "U believes that B believes that no one other than U and B has access to N_B".

$$U \overset{R}{\lhd} N_B \tag{vi}$$

$$U \models B \overset{R}{\mid\!\sim} N_B \tag{vii}$$

$$U \models B \models \{U\}^c \lhd \parallel N_B \tag{viii}$$

$$U \models B \models \{U, B\}^c \lhd \parallel N_B \tag{ix}$$

It is assumed in the protocol that U believes that the base station B is secure and trusted or in other words, we have the inference "U believes that B is the super-principal". which is (x). Applying the ***super-principal rule*** to statements (ix) and (x), we get (xi) which states "U believes that no one other than U and B has access to N_B".

To proceed further we need to understand a few definitions and rules of message idealization from [24] which are as follows:

- If a message does not have any symbols, it is an *atomic message (AM)*.
- An *AM* which is sent in one line and received in another by the origin node is called a *challenge*.
- If a *challenge* is present in a message sent to the origin node, it is a *replied challenge (RC)*.
- A *response* is an *AM* and an *RC* sent together by the node which sends the *response*.
- If an *AM* is neither a *challenge* nor a *response*, it is *nonsense*. All *nonsense* is discarded.
- If an *AM* behaves as a *challenge* as well as a *response* in a line, it is treated as a *response*.
- "*Response* **R** *RC*" is the combination of a *replied challenge* and its *response*.

In the first message of the protocol N_A is sent to B. In response to this, B sends N_B in message 2 which is encrypted in Q. On decryption of Q, U obtains $N_A \| N_B$. Therefore, according to the above mentioned definitions and rules, N_A is a challenge and N_B is the response. Note that these are not the same challenge-response pair (C, R) of the PUF. Thus we have the statement "U can see the replied challenge N_A and the response N_B with decipher key R" which is (xii). Applying the ***intuitive rule*** to (xii), we get (xiii) which is "U can see the replied challenge N_A and the response N_B".

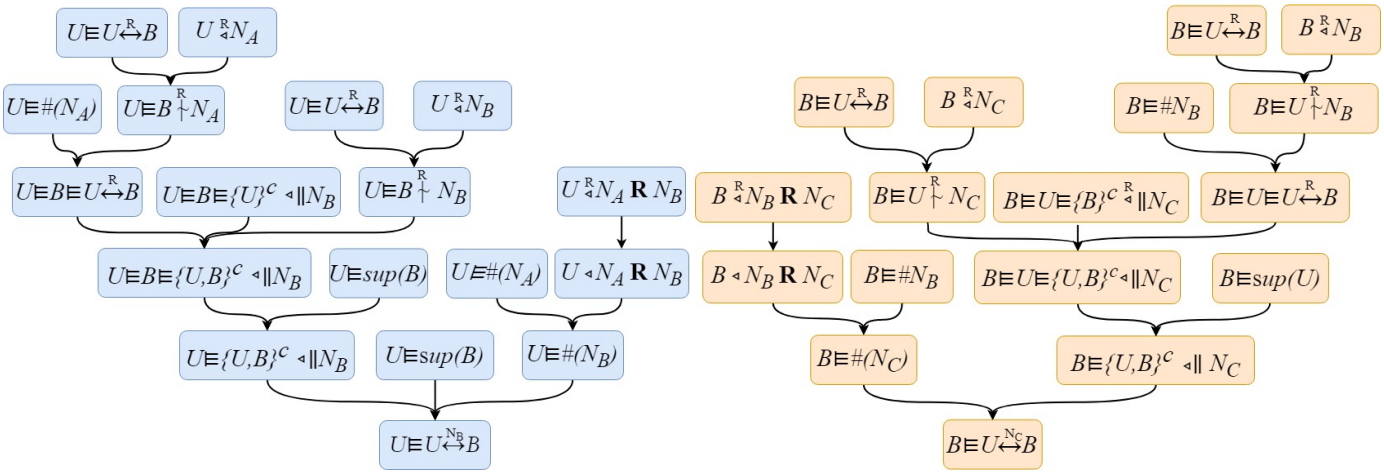
$$U \models sup(B) \tag{x}$$

$$U \models \{U, B\}^c \lhd \parallel N_B \tag{xi}$$

$$U \overset{R}{\lhd} N_A \ \mathbf{R} \ N_B \tag{xii}$$

$$U \lhd N_A \ \mathbf{R} \ N_B \tag{xiii}$$

On applying the ***fresh rule*** to (iv) and (xiii), we get the



(a) U believes that N_B is a good secret known only to U and B

(b) B believes that N_C is a good secret known only to U and B

Fig. 3: Formal Security Proof

statement (xiv) which is "U believes N_B is fresh".

$$U \models \#(N_B) \tag{xiv}$$

$$U \models U \overset{N_B}{\leftrightarrow} B \tag{xv}$$

Applying the ***good-key rule*** to statements (x), (xi) and (xiv) we have proved the statement "U believes that N_B is a good secret known only to U and B". Similarly, we can prove the statement "B believes that N_B is a good secret known only to U and B", as is shown in Fig. 3b. The statements "B believes that N_C is a good secret known only to U and B" and "U believes that N_C is a good secret known only to U and B" are shown in Fig. 4 and Fig. 5 respectively. In these figures, the logical AND operation between two statements is represented by a '$\bigwedge$'. Thus, we have shown that neither N_B nor N_C can be accessed by an adversary. Similarly, it can be proved that an adversary cannot gain access to R'.

If an adversary cannot obtain N_B, N_C and R', it cannot make sense of the communicated data. The Mao Boyd formal proof discussed above has proven the secrecy of N_B, N_C and R' which is regardless of the kind of attack used by the adversary such as masquerade attack, man-in-the-middle (MITM) attack, and replay attack. An adversary cannot masquerade as the drone since it does not have the PUF, and it cannot masquerade as the base station since it does not have the correct challenge-response pairs of any drone. Thus, masquerade and MITM attacks will fail. For the same reason and also because each challenge-response pair is used only once, the proposed scheme is secure against replay attacks. Note that even if an attacker captures the UAV, the property of PUF mentioned in section I ensures that it cannot gain any legitimate (C, R) pair. Additionally, there are no secrets stored on the UAV itself. This guarantees physical security and protection against node tampering attacks. The use of an Alias ID (AID) as in equation (6) guarantees untraceability. Also, a unique session key is generated in each session of authentication.

$$\dfrac{\dfrac{B \models U \overset{R}{\leftrightarrow} B \bigwedge B \models U^c \lhd\| N_B \bigwedge B \overset{R}{\mid\sim} N_B}{B \models \{U,B\}^c \lhd\| N_B} \bigwedge B \models \#(N_B)}{B \models U \overset{N_B}{\leftrightarrow} B}$$

Fig. 4: B believes that N_B is a good secret known only to U and B

$$\dfrac{\dfrac{U \models U \overset{R}{\leftrightarrow} B \bigwedge U \models B^c \lhd\| N_C \bigwedge U \overset{R}{\mid\sim} N_C}{U \models \{U,B\}^c \lhd\| N_C} \bigwedge U \models \#(N_C)}{U \models U \overset{N_C}{\leftrightarrow} B}$$

Fig. 5: U believes that N_C is a good secret known only to U and B

B. Cryptanalysis

This analysis is based on a list of 12 independent security criteria put forward by the authors in [25], which removes redundancies and ambiguities which were prevalent in the area of security protocols. Several works such as [30–33] have already adopted this state-of-the-art security evaluation technique.

1) **[C1] No password verifier table:** In the proposed scheme, the base station stores one challenge-response pair for each registered drone. But, it does not store any user/operator passwords or any derived values of passwords.
2) **[C2] Password friendly:** This criterion is also not applicable to the proposed scheme since users/drone operators are not required to remember any password.
3) **[C3] No password exposure:** In the proposed protocol, the base station server is assumed to be trusted and thus has full access to the database which contains challenge-response pairs of all the registered drones.

TABLE II: Comparison of security features.

Protocols	C5.1	C5.2	C5.3	C5.4	C7	C8	C10	C11	C12
[12]	✗	✗	✓	✗	✗	✓	✗	✗	✗
[13]	✓	✓	✗	✗	✓	✓	✓	✗	✓
[14]	✓	✓	✓	✗	✗	✓	✗	✗	✗
[15]	✗	✓	✓	✓	✓	✗	✓	✓	✗
[16]	✗	✓	✓	✓	✓	✓	✓	✓	✓
[26]	✓	✗	✓	✗	✗	✓	✓	✗	✗
[27]	✗	✓	✓	✓	✗	✗	✓	✗	✗
[28]	✗	✓	✗	✗	✓	✓	✓	✗	✗
[29]	✓	✓	✓	✗	✓	✓	✓	✓	✗
Proposed	✓	✓	✓	✓	✓	✓	✓	✓	✓

4) **[C4] No smart card loss attack:** This criterion is also not applicable to the proposed scheme since no smart cards are required in the considered scenario.
5) **[C5] Resistance to known attacks:** The proposed scheme is secure against masquerade attack, MITM attack, replay attack and node tampering attack. An adversary trying to impersonate as a registered drone cannot possess the same PUF as the drone. Also, it cannot impersonate as the base station because it doesn't have any challenge-response pairs of any of the drones. Thus, the scheme is free from any masquerade attack. Since it can't impersonate either a registered drone or the base station, it is thus free from MITM attack also. Further, there is no possibility of any replay attack since each challenge-response pair is used only once and is discarded in the subsequent round of authentication. Since no secret information related to the authentication process is stored in the drone, the protocol is thus free from node tampering attack. However, it is not secure against de-synchronization attacks [34–36]. If an adversary manages to block the third message of the SecAuthUAV protocol, the database will not be updated with the new challenge-response pair (C', R'). Since the current challenge-response pair (C, R) will not be used more than once, it will be discarded leading to a denial of service.
6) **[C6] Sound repairability:** This criterion is not applicable to the proposed scheme since no smart cards are required in the considered scenario.
7) **[C7] Provision of key agreement:** In the proposed protocol, a new and secure session key Ses_k is established between the drone and the base station after every successful authentication.
8) **[C8] No clock synchronization:** The proposed security scheme is independent of any time delay or clock synchronization, i.e the base station does not need to synchronize its clock with any of the drones and vice versa.
9) **[C9] Timely typo detection:** This criteria is not applicable to the proposed scheme since users/drone operators are not required to remember any passwords.
10) **[C10] Mutual authentication:** The proposed scheme achieves mutual authentication between the drone and the base station. This is enabled by the use of a single PUF challenge-response pair of the drone which is stored on the base station prior to the deployment of the drone. A session key can be established only between a trusted drone and the base station, i.e, if a session key is established, it means that mutual authentication has already been achieved.
11) **[C11] User anonymity:** The use of an Alias ID (AID) as in equation (6) guarantees user anonymity in the proposed scheme. In addition, no message exchange is required to convey the AID to the base station. No entity other than the drone or the base station can know the PUF response R for a given challenge C to compute the subkey K_7 which is required to obtain AID. It has been rigorously proven in the literature that, public-key cryptographic techniques are inherently needed to provide un-traceability and user anonymity [37]. But in this work, user anonymity is achieved with the help of PUFs as discussed above. Although this dynamic ID update mechanism to create AIDs achieves un-traceability, as mentioned in the discussion on C5, it can lead to de-synchronization attacks.
12) **[C12] Forward secrecy:** In the proposed scheme, even if an attacker manages to guess the current session key Ses_k, the security of all future sessions cannot be compromised. This is because the response for the next session, R' cannot be obtained by the adversary as already proven in Section V-A which is required to establish the session key of the next session. Moreover, since the proposed protocol guarantees C11, the adversary will not even be able to track the drone of concern. Thus, the proposed protocol achieves perfect forward secrecy. An adversary will also not be able to get the session key of any previous round either. Thus perfect backward secrecy is also guaranteed by this protocol.

VI. Security and Computational Performance Comparison

A. Security Comparison

This section provides a comparative analysis of our protocol with other existing security schemes available for UAV-BS authentication [12, 13, 26–29]. Table II presents this comparison

based on the 12 criteria discussed in the previous section, In the table, '✓' and '✗' respectively indicate whether a protocol satisfies or does not satisfy a criterion. Criteria **C5.1**, **C5.2**, **C5.3** and **C5.4** refer to protection against Masquerade, MITM, Replay, and Node Tampering attacks respectively. Criteria **C1-C4, C6** and **C9** involve smart cards and user-input passwords which are not applicable to any of the protocols considered for the comparison and hence are not mentioned Table II. As depicted in Table II, our proposed protocol fares well in all these criteria. Most other mentioned protocols provide mutual authentication, while [12] provides one-sided authentication only. For secure authentication and communication, it is desirable to generate a unique session key for each session. Only the protocols in [13, 28, 29] are capable of this. Physical security against node tampering is a very important security metric to be considered for the IoD environments, where the devices like UAVs are deployed in hostile environments and are thus vulnerable to physical tampering and device capture. Other than our proposed protocol, the only other protocol which accounts for the physical security of the individual UAVs is [27]. In [27], upon device capture, the UAV hardware is disabled by sending a destruction signal from the ground station, to protect it from data theft. Further, most of these works only perform some type of informal security analysis on their protocols. For any security protocol, a thorough debugging analysis is crucial to ensure that the protocol is not susceptible to exploitation by malicious entities and also to ensure that vital information is not leaked or altered. Most of the mentioned protocols fail to provide protection against one or more important security attacks as shown in the table. Our protocol is not only safe against the common attacks like masquerading attack, MITM attack, and the replay attack, but also against node tampering attack. Even if an adversary physically captures a UAV, any attempt to tamper with the PUF will render the PUF unusable and therefore the captured UAV will not be able to authenticate with the base station. Thus, the adversary cannot resort to nefarious activities even in the event of a device capture.

B. Simulation and Computational Performance

We used NodeMCU v1.0 and Raspberry Pi 3B as two different simulation environments for the UAV in our system model, to run commonly used mathematical and cryptographic operations such as encryption/decryption, hashing, pseudo-random number generation (PRNG), modular exponent, multiplication, addition, and bitwise XOR. We show a comparison of the total computation time for all the different operations carried out on the above two simulation environments in Fig. 6. From the figure, it can be seen that while [15], [16], [13], [14] took $1.58ms$, $2.30ms$, $1.34ms$ and $3.28ms$ respectively on NodeMCU, the proposed protocol took $1.15ms$ only to execute. Similarly, it can be observed that while [15], [16], [13], [14] took $12.3ms$, $16.70ms$, $29.9ms$ and $41.1ms$ respectively on Raspberry Pi, the proposed protocol took approximately $1.20ms$ to execute. Thus, in terms of computational cost the proposed protocol, SecAuthUAV is shown to perform much better when the NodeMCU platform is used to simulate UAV, while SecAuthUAV performs competitively well in the Raspberry Pi environment. Further, we show that our protocol is scalable even when the value of 'm' is increased. Fig. 7 shows the computational cost with increasing values of 'm'. It can be observed that even when the value of 'm' goes up to 1000, the total time taken or the computational cost increases only slightly.

We further do a comparison of the proposed protocol with other protocols in terms of communication cost. Standard sizes for the different fields communicated across the entities are chosen similar to [15], [16], where the timestamp is 32 bits long; nonce is 128 bits long; while hash, MAC and device ID are 160 bits long each. As shown in Table III, based on these sizes the communication overhead in [15], [16] and [13] is 1024 bits, 864 bits, and 1184 bits respectively, while in our protocol the communication overhead is just 704 bits.

We also do a comparison of the proposed protocol with other protocols in terms of storage cost in number of bits for storing various data fields in the UAV's memory. Standard sizes for the different data fields have been considered for this

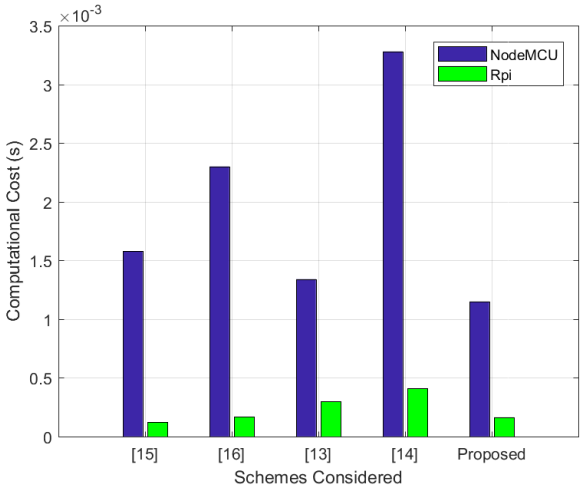


Fig. 6: Computational cost comparison.

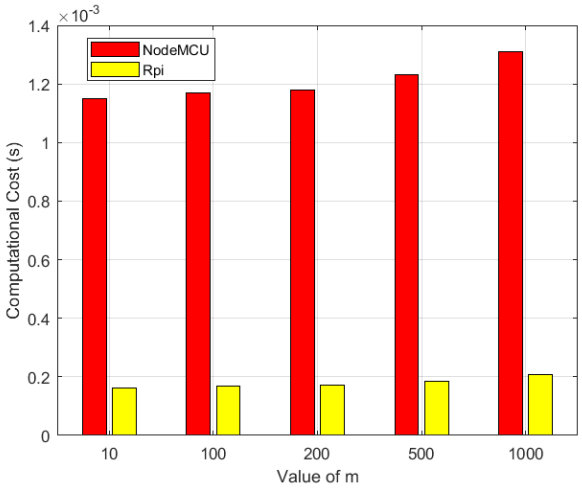


Fig. 7: Computational cost of SecAuthUAV by varying the value of 'm'.

TABLE III: Comparison of communication cost in bits.

	Protocols			
	[15]	**[16]**	**[13]**	**Proposed**
Communication cost	1024 bits	864 bits	1184 bits	704 bits

TABLE IV: Comparison of storage cost in bits.

	Protocols			
	[15]	**[16]**	**[13]**	**Proposed**
Storage cost	480* bits	480* bits	448 bits	160 bits

$*$ 480 bits is the minimum storage in [15],[16]

comparison, where the timestamp is 32 bits long; device ID and temporary IDs are 160 bits long, and the private key is at least 256 bits long. As shown in Table IV, the memory storage sizes in [15], [16] and [13] are at least 480 bits, 480 bits, and 448 bits long respectively, while in our protocol the memory storage cost is just 160 bits since only the device ID of the UAV needs to be stored in its memory.

Therefore, our scheme is a better choice compared to the existing security schemes for UAVs considering both the computational capabilities of UAVs and the communication costs involved.

VII. Conclusions

This paper proposed SecAuthUAV, a lightweight mutual authentication protocol for UAV-BS authentication using physically unclonable functions (PUFs). SecAuthUAV uses a challenge-response scheme enabled by PUFs which gives it the advantage of not having to store any secret information in the UAV nodes. Only one challenge-response pair per UAV is stored in the base station at a time. In the process of authenticating with the base station, a secure session key is established between them using the secret information generated using PUFs. We showed that SecAuthUAV ensures mutual authentication, un-traceability, and physical security along with protection against various attacks such as masquerade, MITM, replay, and node tampering attacks. It is also capable of generating a unique session key for each new session. Moreover, SecAuthUAV fares well in comparison with the other existing authentication solutions for UAVs in terms of computational, communication and memory storage cost. Hence, we argue that the proposed protocol, SecAuthUAV is very efficient and viable for UAV-BS authentication.

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